

CS193

Homework - Week 7

Introduction

In lecture you learned how to go beyond typing one command at a time and start **scripting** in bash. This week you'll put those skills to work by writing a small but genuinely useful tool: a **study-log helper**. Each time you run it, it records a timestamped study session for you and reports how many sessions you've logged so far.

You only need **bash and a terminal** — no other languages, no installs.

We are giving you a working starter template. You are not writing the whole script from scratch. The scaffold is laid out with **TODO** comments, and each one points you to the exact lecture slide that shows the syntax you need. The goal is to leave with a script that runs and that cements variables, arguments, test operators, redirection, and command substitution. It should take about **15–20 minutes**.

Getting Started: The Template

Open the `hw7-studylog` folder. It already has the structure and hints you need — you just fill in the **TODO** sections.

- `studylog.sh` — the script you'll write. Look for **# TODO** comments. Each one tells you what to add and names the lecture slide with the matching syntax.
- `README.md` — a quick reminder of how to make the script executable and run it.
- `.vscode/` — recommended extensions and settings. You don't need to touch this; it just makes editing shell scripts nicer.
- `study_log.txt` — you won't see this at first. Your script *creates* it the first time you run it.

How to Run Your Script

Using VS Code

1. Open VS Code and go to **File** → **Open Folder**, then select the `hw7-studylog` folder.
2. Open a terminal inside VS Code with **Terminal** → **New Terminal**. It opens already pointed at your project folder.
3. Make the script executable (one time), then run it, passing your name:

```
$ chmod +x studylog.sh
$ ./studylog.sh Boiler
Thanks Boiler! You have logged 1 study session(s).
```

4. Run it a few more times and watch the count grow. Open `study_log.txt` to see your entries.

5. Not using VS Code? The same idea applies in any terminal: `cd` into the `hw7-studylog` folder, then run the two commands above. No server, compiler, or installation is required — it's plain bash.

What to Fill In

Work through the TODO comments in `studylog.sh` in order:

1. **Shebang.** Keep `#!/bin/bash` as the very first line (already done for you). It tells the OS which interpreter to use.
2. **Check for a name.** If the student runs the script with no argument, print a usage message and **exit with a non-zero code**. Hint: test with `-z "$1"` inside an `if`.
3. **Use a variable.** Store the log file name in a variable, e.g. `LOGFILE="study_log.txt"`. Remember: **no spaces** around the `=`.
4. **Append a timestamped entry.** Add a line such as `<name> studied on <date>` to the log using `$1`, command substitution `$(date)`, and `>>` (append, **not** `>`, which would erase the file each run).
5. **Count and report.** Use `wc -l` to count the lines in the log, then print a message with the total.

Optional stretch (not required for full credit): add an `if` with the `-ge` test operator that prints an encouraging message once you've logged 3 or more sessions, and a `for` loop that lists your existing entries.

How to Submit

1. Run your script at least twice so `study_log.txt` has entries in it.
2. Zip your `hw7-studylog` folder, with your finished `studylog.sh` and the `study_log.txt` it produced included:
 - **Windows:** Right-click the `hw7-studylog` folder → **Send to** → **Compressed (zipped) folder**.
 - **Mac:** Right-click the `hw7-studylog` folder → **Compress** "`hw7-studylog`".
3. Go to Ed Discussion and open the **Week 7** submission thread.
4. Create a new post, attach your zipped folder, and submit.

Checklist Before You're Done

- `#!/bin/bash` is still the first line of the script - 10%
- Running with no name prints a usage message and exits non-zero - 20%
- The log file name is stored in a variable (no spaces around `=`) - 15%

- A timestamped entry is appended using `$1`, `$(date)`, and `>>` - 25%
- The total number of sessions is counted with `wc -l` and printed - 20%
- You ran the script at least twice and `study_log.txt` shows your entries - 10%

Grading

This assignment is graded on completion of the checklist above. Each item is worth the percentage listed next to it, and the items add up to 100% of your grade on this assignment.